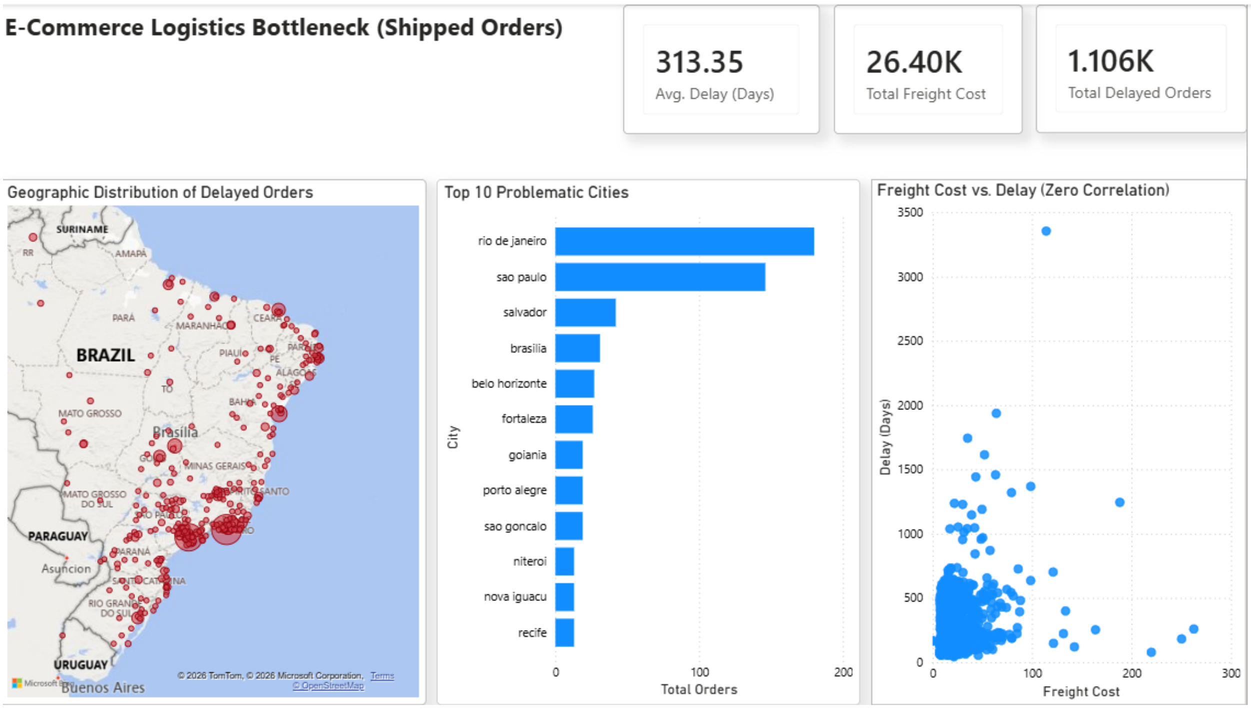
Project: E-Commerce Supply Chain Bottleneck Analysis (Olist Dataset)



Executive Summary & Insights:

• **The Challenge:** We analyzed an e-commerce dataset to understand why a large volume of orders missed their delivery dates. The goal was to find the exact problem: is it the internal warehouse or the external delivery carriers?

• **Key Insight 1 (Fast Warehouse):** The internal warehouse is not the problem. It works very well, taking an average of only **3.3 days** to process orders and give them to the carriers.

• **Key Insight 2 (Carrier Delays):** The external carriers are the main cause of the problem. They kept **1,107** "shipped" orders for an average of **311 days** without delivering them to the customers.

• **Key Insight 3 (Cost vs. Speed):** Data showed no connection between high shipping costs and faster delivery. Paying more for shipping does not stop carrier delays.

Business Recommendations:

• **Recommendation 1:** Review the contracts (SLAs) with the current delivery carriers. Hold them accountable for the 311-day delays, ask for financial compensation, or find new, reliable logistics partners.

• **Recommendation 2:** Contact the 1,107 affected customers immediately. Offer them refunds or send new products to keep their trust in the brand.

Business Impact: Fixing carrier contracts and focusing on cities with the most delays (like Rio de Janeiro and Sao Paulo) will reduce lost customers and improve overall delivery success.

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Technical Execution & Data Pipeline:

1.Methodology & Data Process:

To find the root cause of the delivery delays, I used Google BigQuery (SQL) to clean and analyze the data. I used date functions (TIMESTAMP_DIFF) and a subquery to find the most recent date in the database to calculate exactly how many days the orders were overdue. After joining the data from multiple tables, I connected the results to Power BI to visualize where the bottlenecks were happening

2.The SQL Code (Date Calculations & Joins):

Below is a snippet of the SQL query I wrote to calculate the carrier delay and check if there is a relationship between shipping costs and delivery speed.

```sql
-- Q7: Correlation between Freight Value (Shipping Cost) and Carrier Delay.
-- Insight: Does paying premium shipping prevent carrier delays?
SELECT
    c.customer_city,
    ROUND(AVG(oi.freight_value), 2) AS avg_freight_paid,
    ROUND(AVG(TIMESTAMP_DIFF(
        (SELECT MAX(order_purchase_timestamp)
         FROM `data-analysis-projects-496119.olist_delivery_analysis.orders`),
        o.order_delivered_carrier_date, DAY)), 1) AS avg_carrier_delay
FROM `data-analysis-projects-496119.olist_delivery_analysis.orders` o
JOIN `data-analysis-projects-496119.olist_delivery_analysis.customers` c
    ON o.customer_id = c.customer_id
JOIN `data-analysis-projects-496119.olist_delivery_analysis.order_items` oi
    ON o.order_id = oi.order_id
WHERE o.order_delivered_customer_date IS NULL
  AND o.order_status = 'shipped'
GROUP BY c.customer_city
ORDER BY avg_carrier_delay DESC;
```

The complete SQL scripts, including complex joins, subqueries, and categorizations (CASE WHEN), are available on my GitHub.

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